

lab4

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1 clear;
2 clc;
3 close all;
4 pkg load queueing
5 format long g
6
7 % Lab 4 - Erlang C
8 % Minimal extension of the provided helper code
9
10 lambda = 100;           % Calls per hour
11 h_minutes = 3;          % Mean call duration [minutes]
12 h = h_minutes / 60;     % Mean call duration [hours]
13 mu = 1 / h;             % Calls per server per hour = 20
14 m = 10;                 % Number of agents / servers
15
16 % Service Level threshold t
17 % The statement leaves it as "t".
18 % Default choice here: 20 sec (typical 80/20 target from the intro slides).
19 % If your instructor wants another value, change only this line.
20 t_seconds = 20;
21 t = t_seconds / 3600;   % [hours]
22
23 % a) Offered load
24 A = lambda * h;         % Equivalent to lambda / mu
25
26 % Erlang C stability condition
27 if A >= m
28     error('Unstable Erlang-C system: A must be smaller than m.');
```

$$P_0 = \frac{1}{\sum_{k=0}^{m-1} \frac{A^k}{k!} + \frac{A^m}{m!} (1 - \frac{A}{m})}$$

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29 endif
30
31 % b) P0
32 sum_terms = 0;
33 for k = 0:(m - 1)
34     sum_terms = sum_terms + (A^k) / factorial(k);
35 endfor
36 tail_term = ((A^m) / factorial(m)) * (m / (m - A));
37 P0 = 1 / (sum_terms + tail_term);
38
39 % c) Probability of waiting
40 Pw = erlangc(A, m);      % From queueing package
41 Pw_formula = tail_term * P0; % Closed-form cross-check
42
43 % d) Service Level
44 %  $SL(t) = 1 - Pw * \exp(-(m * \mu - \lambda) * t)$ 
45 SL = 1 - Pw * exp(-(m * mu - lambda) * t);
46
47 % e) Average Speed of Answer
48 ASA_hours = (Pw * h) / (m - A);
49 ASA_minutes = ASA_hours * 60;
50 ASA_seconds = ASA_hours * 3600;
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51
52 % Extra sanity metric
53 rho = A / m; % Occupancy / utilization
54
55 fprintf('===== Erlang C - Lab 4 =====\n');
56 fprintf('lambda          = %.10f calls/hour\n', lambda);
57 fprintf('h              = %.10f minutes/call\n', h_minutes);
58 fprintf('mu              = %.10f calls/server/hour\n', mu);
59 fprintf('m              = %d servers\n', m);
60 fprintf('t              = %.10f seconds\n', t_seconds);
61 fprintf('-----\n');
62 fprintf('(a) Offered load A      = %.10f Erlangs\n', A);
63 fprintf('(b) P0              = %.10f\n', P0);
64 fprintf('(c) Pw (erlangc)       = %.10f\n', Pw);
65 fprintf('    Pw (closed-form)    = %.10f\n', Pw_formula);
66 fprintf('    abs difference      = %.10e\n', abs(Pw - Pw_formula));
67 fprintf('(d) Service Level SL(t) = %.10f\n', SL);
68 fprintf('                    = %.10f %%\n', 100 * SL);
69 fprintf('(e) ASA              = %.10f hours\n', ASA_hours);
70 fprintf('                    = %.10f minutes\n', ASA_minutes);
71 fprintf('                    = %.10f seconds\n', ASA_seconds);
72 fprintf('Extra: rho = A / m      = %.10f\n', rho);
73 fprintf('                    = %.10f %%\n', 100 * rho);
74 fprintf('===== \n');

```